Assignment 8 — Task List, Milestones, Timeline, and Effort Matrix

CS 5001 / CS 5002 Senior Design Project — Fall 2026 / Spring 2027

Project: Smart Closet Mirror (Personal Stylist)
Team: Bull riders
Team Members: Rehman Khalid, Arya Garg, Jace Schubert, Clare Hooper
Faculty Advisor: Hrishikesh Bhide
Due Date: October 28, 2026
Implementation Horizon: January 11, 2027 – April 01, 2027 (Expo: April 2027)

1. Component 1: Task List (Assignment #5)

Twenty project-specific numbered tasks, exactly five per team member, each sized to 1–2 weeks and beginning with an approved verb naming the exact Design Diagram subsystem/module. Saved as `Tasklist.md` in the repository root.

Rehman Khalid (L1 Capture, Planes, Infrastructure)

1. **T01.** `Develop` the **L1 Capture** Kinect frame acquisition daemon with USB error recovery and shared-memory ring buffer (I1, I7). Owner: Rehman Khalid.
2. **T02.** `Develop` the **Planes Message Bus and Data Store** Redis broker and SQLite schema migrations for state and telemetry. Owner: Rehman Khalid.
3. **T03.** `Develop` the **L1 Capture** privacy consent hardware gate and camera status beacon integration (I27). Owner: Rehman Khalid.
4. **T04.** `Develop` the **Planes Logging and Health** watchdog service for layer heartbeats, crash restarts, and drop-rate telemetry. Owner: Rehman Khalid.
5. **T05.** `Validate` the **L1 Capture and Planes** pipeline for sustained 30 fps throughput under 50 ms capture-to-shm latency. Owner: Rehman Khalid.

Arya Garg (L2 Perception, L3 Body, L5 Try-on)

1. **T06.** `Develop` the **L2 Perception** body joint detection and T-pose extraction pipeline using depth and RGB streams (I7, I8). Owner: Arya Garg.
2. **T07.** `Develop` the **L3 Body** avatar generator translating measurement vectors into rigged 3D avatar meshes (I8, I9). Owner: Arya Garg.
3. **T08.** `Develop` the **L5 Try-on** fit evaluation engine calculating garment tension and size compatibility (I11, I12). Owner: Arya Garg.
4. **T09.** `Develop` the **L5 Try-on** live garment overlay and 2D pose-driven warp video pipeline (A2, I10). Owner: Arya Garg.
5. **T10.** `Validate` the **L2 Perception and L5 Try-on** H.264 video stream encoder against the 250 ms motion-to-glass latency budget (I4). Owner: Arya Garg.

Jace Schubert (L4 Wardrobe, L7 Connectors, Glue Auth)

1. **T11.** `Develop` the **L4 Wardrobe** REST ingestion service and image storage for phone-submitted garment assets (I2). Owner: Jace Schubert.
2. **T12.** `Develop` the **L7 Connectors** calendar and weather normalization adapter with token refresh and stale-cache fallback (I3, I14). Owner: Jace Schubert.
3. **T13.** `Develop` the **L7 Connectors** product catalog search and external image fetcher with rate limiting (I3, I14). Owner: Jace Schubert.
4. **T14.** `Develop` the **Glue: Authentication** 6-digit phone-pairing handshake and session authorization handler. Owner: Jace Schubert.
5. **T15.** `Test` the **L4 Wardrobe and L7 Connectors** offline resilience and data plane consistency under simulated network failures. Owner: Jace Schubert.

Clare Hooper (L6 Decision, L8 Presentation Lead, Glue Notifications)

1. **T16.** `Develop` the **L6 Decision** outfit recommendation scoring engine matching closet items to weather and calendar context (I13, I14). Owner: Clare Hooper.
2. **T17.** `Develop` the **L6 Decision** buy or skip scoring algorithm and wardrobe gap analyzer (I12, I14). Owner: Clare Hooper.
3. **T18.** `Develop` the **L8 Presentation** WebView UI deck and decision card rendering engine compatible with WebView 44 (I6, I17). Owner: Clare Hooper.
4. **T19.** `Develop` the **Glue: Notifications** toast broadcast service and Mirror presence/wake controller (I5, I6). Owner: Clare Hooper.

5. **T20.** `Validate` the **L6 Decision and L8 Presentation** end-to-end user recommendation flow against gesture and voice action timeouts. Owner: Clare Hooper.

2. Component 2: Milestones List (Assignment #6)

Every major planned milestone and completed deliverable with detailed explanation: what it demonstrates, date, verification method, and verifier. Completed items are marked. Implementation milestones are spaced every 1–3 weeks.

Milestone	Target Date	Detailed Outcome & Demonstration	Verification Method	Verifier
M01 (Done)	Sept 02, 2026	**Professional Biographies:** Team member backgrounds, technical competencies, and role allocations established.	Canvas submission & repo commit.	Course Instructor
M02 (Done)	Sept 09, 2026	**Team Formation & Project Description:** Core project scope, problem statement, and team contract approved.	Advisor acceptance & contract.	H. Bhide (Advisor)
M03 (Done)	Sept 16, 2026	**Constraints Essay:** Technical, economic, ethical, and privacy constraints analyzed (camera consent, weak Android hardware).	Graded essay evaluation.	Course Instructor
M04 (Done)	Sept 23, 2026	**User Stories & Use Cases:** Stakeholder map, user stories (US-01–US-04), and error handling flows baselined.	User_Stories.md review.	Course Instructor
M05 (Done)	Oct 07, 2026	**High-Level Design (D0):** L1–L8 layer boundaries, component responsibility matrix, and interface catalog (I1–I17).	Peer review & rubrics.	Peer Reviewers
M06 (Done)	Oct 14, 2026	**Detailed Design (D1):** Relational schema, indexes, and core algorithms (A1 fit calculation, A2 live pose warp).	D1 report & data model diagram.	Course Instructor
M07 (Done)	Oct 21, 2026	**Detailed Design (D2):** Sequence diagrams, consent state machine (I27), and peer review action items.	Peer review action log.	Course Instructor
M08 (Done)	Oct 28, 2026	**Project Plan & Baseline (A8):** Task list, milestones, Gantt timeline, effort matrix, and risk register baselined.	Consolidated PDF & Tasklist.md.	Course Instructor
M09	Dec 11, 2026	**Design Freeze & Dev Environment Verification:** Toolchains, Ubuntu PC OS, Kinect SDK, and Mirror shell harness ready.	Local smoke test & driver check.	Rehman Khalid
M10	**Jan 22, 2027**	**Sensor Ingest & Data Plane Skeleton: L1 publishes 30 fps FrameBundles to shm; L4 accepts garment images via REST.**	**Unit tests; capture.health confirms 30 fps stream without drops.**	**Rehman Khalid & Jace Schubert**
M11	**Feb 05, 2027**	**Perception & Context Normalization MVP: L2 extracts body dimensions from T-pose (I8); L7 normalizes calendar/weather data.**	**Test scripts verify cm-accurate joints and valid context payloads.**	**Arya Garg & Jace Schubert**
M12	**Feb 19, 2027**	**Body Avatar & Recommendation Core: L3 outputs rigged glTF mesh (I9); L6 outputs weather-matched outfit rankings.**	**Benchmark tests verify outfit rankings; avatar renders in glTF viewer.**	**Clare Hooper & Arya Garg**
M13	**Mar 05, 2027**	**Try-on Stream & Glass Presentation Alpha: L5 streams H.264 video to Media3; L8 WebView displays live decision cards.**	**Visual check on mirror glass; hardware decode confirmed on Android 6.**	**Clare Hooper & Rehman Khalid**
M14	**Mar 26, 2027**	**Subsystem Integration & Latency Gate: Complete sensor-to-glass pipeline runs within 250 ms motion budget; offline fallback verified.**	**Telemetry logs confirm latency ≤ 250 ms across 100 continuous frames.**	**Entire Team**
M15	**Apr 01, 2027**	**Final Project Completion & Acceptance Freeze: End-to-end journey (pairing, sizing, outfit choice, buy/skip) verified against requirements.**	**Formal acceptance sign-off; code freeze committed to repository.**	**H. Bhide (Advisor) & Instructor**
M16	**Apr 15, 2027**	**Senior Design II Expo Demonstration: Continuous, error-free demonstration during public Senior Design Expo evaluation.**	**Live public demonstration & judging rubric score.**	**Expo Evaluators**

3. Component 3: Table 1 — Timeline (Assignment #6)

Every Task List item and milestone completion date with dependencies, explicit 15–20% buffer per task, protected pre-milestone blocks, and schedule risk visibility. Implementation horizon: Jan 11 – Apr 01, 2027.

ID	Task / Milestone Statement	Owner	Start	End	Dependencies	Buffer & Risk Visibility
T01	Develop L1 Kinect acquisition daemon & shm ring buffer	Rehman	2027-01-11	2027-01-22	None	2 days (20%). Mitigates R05.
T02	Develop Planes Message Bus & Data Store (Redis/SQLite)	Rehman	2027-01-11	2027-01-22	None	2 days (20%).
T11	Develop L4 Wardrobe REST ingestion & image storage	Jace	2027-01-11	2027-01-22	None	2 days (20%).
T14	Develop Glue: Auth 6-digit phone-pairing handshake	Jace	2027-01-11	2027-01-22	None	2 days (20%).
M10	**MILESTONE: Sensor Ingest & Data Plane Skeleton**	**Rehman, Jace**	**2027-01-22**	**2027-01-22**	**T01, T02, T11, T14**	**Protected review checkpoint.**
T03	Develop L1 Capture privacy consent hardware gate (I27)	Rehman	2027-01-25	2027-02-05	T01, T02	2 days (20%).
T06	Develop L2 Perception body joint detection & T-pose	Arya	2027-01-25	2027-02-05	T01	2 days (20%). Mitigates R06.
T12	Develop L7 Calendar/weather normalization adapter	Jace	2027-01-25	2027-02-05	T02	2 days (20%). Mitigates R03.
M11	**MILESTONE: Perception & Context Normalization MVP**	**Arya, Jace**	**2027-02-05**	**2027-02-05**	**T03, T06, T12**	**Protected review checkpoint.**
T07	Develop L3 Body avatar rigged mesh generator (glTF)	Arya	2027-02-08	2027-02-19	T06	2 days (20%).
T13	Develop L7 Product catalog search & image fetcher	Jace	2027-02-08	2027-02-19	T12	2 days (20%).
T16	Develop L6 Recommendation scoring engine	Clare	2027-02-08	2027-02-19	T02, T12	2 days (20%).
M12	**MILESTONE: Body Avatar & Recommendation Core**	**Clare, Arya**	**2027-02-19**	**2027-02-19**	**T07, T13, T16**	**Protected review checkpoint.**
T08	Develop L5 Try-on fit evaluation & tension mapping	Arya	2027-02-22	2027-03-05	T07	2 days (20%).
T09	Develop L5 Live garment overlay & 2D warp pipeline	Arya	2027-02-22	2027-03-05	T08	2 days (20%). Mitigates R02.
T17	Develop L6 Buy/skip decision algorithm & gap analyzer	Clare	2027-02-22	2027-03-05	T16	2 days (20%).
T18	Develop L8 Presentation WebView UI deck (WebView 44)	Clare	2027-02-22	2027-03-05	T16	2 days (20%). Mitigates R01.
M13	**MILESTONE: Try-on Stream & Presentation Alpha**	**Clare, Rehman**	**2027-03-05**	**2027-03-05**	**T08, T09, T17, T18**	**Protected review checkpoint.**
T04	Develop Planes Watchdog heartbeat & crash restart	Rehman	2027-03-08	2027-03-12	T02	1 day (20%).
T19	Develop Glue: Notifications toast broadcast & presence	Clare	2027-03-08	2027-03-12	T02, T18	1 day (20%).
RECESS	*SPRING BREAK (University Holiday — Scheduled Recess)*	*Team*	*2027-03-15*	*2027-03-21*	*None*	*Mitigates Risk R04 (Fatigue).*
T05	Validate L1/Planes sustained 30 fps throughput	Rehman	2027-03-22	2027-03-26	T01, T03, T04	1 day (20%).
T10	Validate L2/L5 250 ms motion-to-glass video latency	Arya	2027-03-22	2027-03-26	T09	1 day (20%). Mitigates R02.
T15	Test L4/L7 offline resilience & consistency fallback	Jace	2027-03-22	2027-03-26	T11, T13	1 day (20%). Mitigates R03.

ID	Task / Milestone Statement	Owner	Start	End	Dependencies	Buffer & Risk Visibility
T20	Validate L6/L8 end-to-end decision flow & timeouts	Clare	2027-03-22	2027-03-26	T17, T18	1 day (20%).
M14	**MILESTONE: Subsystem Integration & Latency Gate**	**Entire Team**	**2027-03-26**	**2027-03-26**	**T05, T10, T15, T20**	**Protected gate before freeze.**
M15	**MILESTONE: Final Project Completion & Acceptance Freeze**	**Advisor & Team**	**2027-04-01**	**2027-04-01**	**M14**	**Complete project sign-off.**
M16	**MILESTONE: Senior Design II Expo Demonstration**	**Expo Evaluators**	**2027-04-15**	**2027-04-15**	**M15**	**Live Public Expo Judging.**

4. Component 4: Table 2 — Effort Matrix (Assignment #6)

Division of labor in estimated hours per task and team member. The primary owner carries the largest share (≥65%) of each task. Estimates derived via PERT formula (O + 4M + P) / 6. Row and column totals are included with a rebalancing plan.

ID	Task Statement	Rehman	Arya	Jace	Clare	Total	Primary Owner
T01	Develop L1 Capture frame acquisition daemon	**26**	4	2	0	32	Rehman Khalid
T02	Develop Planes Message Bus & Data Store	**24**	2	4	2	32	Rehman Khalid
T03	Develop L1 Capture privacy consent hardware gate	**20**	0	2	4	26	Rehman Khalid
T04	Develop Planes Logging & Health watchdog service	**22**	2	2	2	28	Rehman Khalid
T05	Validate L1 Capture & Planes 30 fps throughput	**22**	4	2	0	28	Rehman Khalid
T06	Develop L2 Perception joint detection & T-pose	4	**26**	0	0	30	Arya Garg
T07	Develop L3 Body avatar rigged mesh generator	0	**26**	0	2	28	Arya Garg
T08	Develop L5 Try-on fit evaluation engine	0	**24**	2	4	30	Arya Garg
T09	Develop L5 Try-on live garment overlay & warp	2	**28**	0	2	32	Arya Garg
T10	Validate L2/L5 H.264 video stream 250ms latency	4	**22**	0	2	28	Arya Garg
T11	Develop L4 Wardrobe REST ingestion & image store	2	0	**26**	2	30	Jace Schubert
T12	Develop L7 Connectors calendar/weather adapter	0	0	**24**	4	28	Jace Schubert
T13	Develop L7 Connectors catalog search & fetcher	0	0	**24**	4	28	Jace Schubert
T14	Develop Glue: Auth phone-pairing handshake	2	0	**22**	2	26	Jace Schubert
T15	Test L4/L7 offline resilience & consistency fallback	2	0	**22**	2	26	Jace Schubert
T16	Develop L6 Decision outfit recommendation scoring	0	2	4	**26**	32	Clare Hooper
T17	Develop L6 Decision buy/skip algorithm & gap analyzer	0	2	2	**24**	28	Clare Hooper
T18	Develop L8 Presentation WebView UI deck & cards	0	0	4	**28**	32	Clare Hooper
T19	Develop Glue: Notifications toast & presence	2	0	2	**22**	26	Clare Hooper
T20	Validate L6/L8 end-to-end decision flow & timeouts	2	2	2	**22**	28	Clare Hooper
COLUMN TOTALS (Hours)		**132**	**140**	**140**	**150**	**562**	**Balanced**

Workload Analysis & Rebalancing Plan:
The expected division of labor totals 562 person-hours across the 11.5-week implementation semester (averaging ~12 hours per week per student). The variance between member columns is tightly bounded within a 12% margin (132 to 150 hours). Clare's slightly higher load reflects the presentation layer integration on the physical mirror glass. In the event that WebView 44 styling or polyfill overhead creates a bottleneck, Jace Schubert will assume 10 hours of presentation card template binding (T18), bringing Clare to 140 hours and Jace to 150 hours, thereby preserving equitable team distribution.

5. Supporting Artifact: Work Breakdown Structure (WBS)

Two-level hierarchical breakdown by project phase and task, mapping every Task List item into its system architecture stage.

```
1.0 Project Infrastructure & Sensor Ingestion (Sprint 1: Jan 11 – Jan 22, 2027)
   1.1 T01: Develop L1 Capture Kinect acquisition daemon & shm ring buffer [Rehman Khalid]
   1.2 T02: Develop Planes Message Bus & Data Store (Redis/SQLite) [Rehman Khalid]
   1.3 T03: Develop L1 Capture privacy consent hardware gate & camera beacon [Rehman Khalid]
   1.4 T11: Develop L4 Wardrobe REST ingestion service & image store [Jace Schubert]
   1.5 T14: Develop Glue Authentication 6-digit phone-pairing handshake [Jace Schubert]

2.0 Perception & Normalization Pipeline (Sprint 2: Jan 25 – Feb 05, 2027)
   2.1 T06: Develop L2 Perception body joint detection & T-pose extraction [Arya Garg]
   2.2 T07: Develop L3 Body avatar generation & rigged glTF mesh pipeline [Arya Garg]
   2.3 T12: Develop L7 Connectors calendar & weather normalization adapter [Jace Schubert]
   2.4 T13: Develop L7 Connectors product catalog search & image fetcher [Jace Schubert]

3.0 Decision & Virtual Try-On Engine (Sprint 3 & 4: Feb 08 – Mar 05, 2027)
   3.1 T08: Develop L5 Try-on fit evaluation engine (tension mapping) [Arya Garg]
   3.2 T09: Develop L5 Try-on live garment overlay & 2D pose-driven warp video [Arya Garg]
   3.3 T16: Develop L6 Decision outfit recommendation scoring engine [Clare Hooper]
   3.4 T17: Develop L6 Decision buy/skip scoring algorithm & wardrobe gap analyzer [Clare Hooper]
   3.5 T19: Develop Glue Notifications toast broadcast & presence wakeup [Clare Hooper]

4.0 Presentation & Thin-Client Integration (Sprint 4 & 5: Feb 22 – Mar 12, 2027)
   4.1 T18: Develop L8 Presentation WebView UI deck & decision cards [Clare Hooper]
   4.2 T04: Develop Planes Logging & Health watchdog heartbeat service [Rehman Khalid]

5.0 Verification, Hardening & Acceptance (Sprint 6 & 7: Mar 22 – Apr 01, 2027)
   5.1 T05: Validate L1 Capture & Planes sustained 30 fps throughput [Rehman Khalid]
   5.2 T10: Validate L2 Perception & L5 Try-on 250 ms motion-to-glass latency [Arya Garg]
   5.3 T15: Test L4 Wardrobe & L7 Connectors offline resilience & fallback [Jace Schubert]
   5.4 T20: Validate L6 Decision & L8 Presentation end-to-end decision flow [Clare Hooper]
```

6. Supporting Artifact: Risk Register

Risks across technical, team, and external categories, with explicit likelihood, impact, strategy, mitigation, owner, and trigger. High-impact schedule risks are cross-referenced in Table 1.

ID	Category	Risk Description	Likelihood	Impact	Strategy	Mitigation Plan	Owner	Trigger
R01	Technical	**WebView 44 Incompatibility:** Obsolete Android 6 browser engine rejects ES6/modern CSS, crashing mirror shell UI.	High	High	Reduce	Enforce strict Babel ES5 transpile; avoid CSS grid; test all cards in Chromium 44 emulator before mirror deployment (buffered in T18).	Clare	Logcat syntax error or blank card on device.
R02	Technical	**Motion-to-Glass Latency > 250 ms:** Pose estimation, mesh warping, and H.264 encode create visible mirror lag.	Medium	High	Reduce	Use zero-copy shared memory (I7); downscale stream to 480p or fall back to 2D affine warp mode if frame processing > 20 ms (buffered in T09, T10).	Arya	Watchdog reports pipeline latency > 220 ms.
R03	External	**Third-Party API Outages** / **Throttling:** Google Calendar or weather APIs throttle requests or suffer outages.	Medium	Medium	Avoid	Implement L7 stale-while-revalidate local caching (24h TTL); fall back to default seasonal context bundles (buffered in T12, T15).	Jace	L7 logs 3 consecutive HTTP 429/5xx responses.
R04	Team	**Academic Workload Spikes:** Midterm exam periods or project deliverables cause sprint handoff bottlenecks.	High	Medium	Reduce	Incorporate explicit 15–20% buffer on all tasks; lock Spring Break as zero-task recess; cross-train subsystem pairings.	Team	Any task slips > 4 calendar days past start date.

ID	Category	Risk Description	Likelihood	Impact	Strategy	Mitigation Plan	Owner	Trigger
R05	Technical	**Kinect USB Isochronous Packet Drops:** Linux USB 2.0 driver drops packets under CPU spikes, stalling pipeline.	Low	High	Reduce	Assign real-time SCHED_FIFO priority to L1 daemon; automated 2-second USB re-enumeration fallback (buffered in T01).	Rehman	L1 emits capture.health = DOWN for > 2 s.
R06	External	**Kinect Bracket Mechanical Drift:** 3D-printed clip bracket flexes or slips, distorting cm ground-truth sizing.	Low	Medium	Avoid	Print bracket with stiffening gusset in PETG; run 10-second reference plane calibration check during boot (buffered in T06).	Rehman	Extracted user height deviates > 10% across runs.